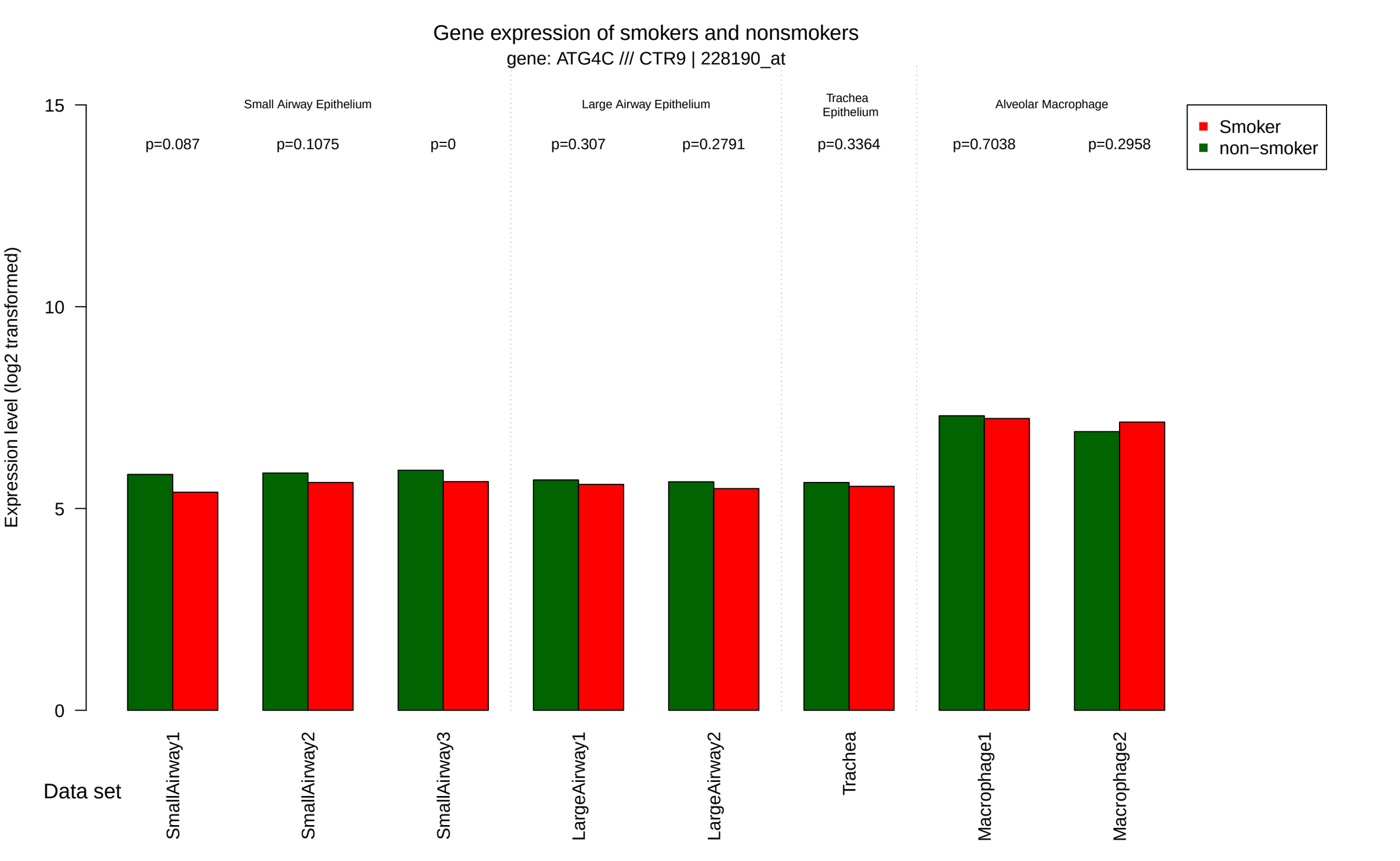
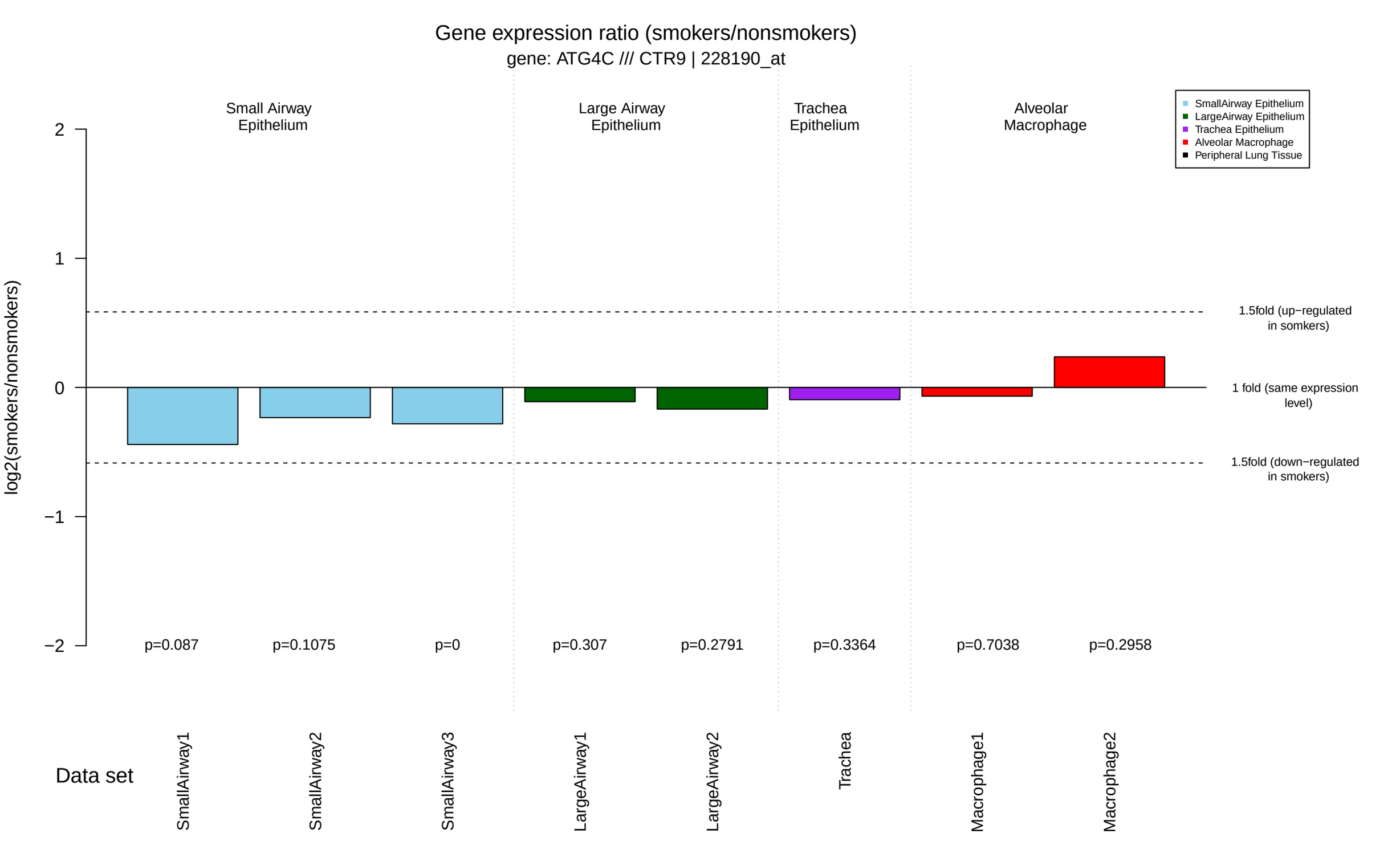
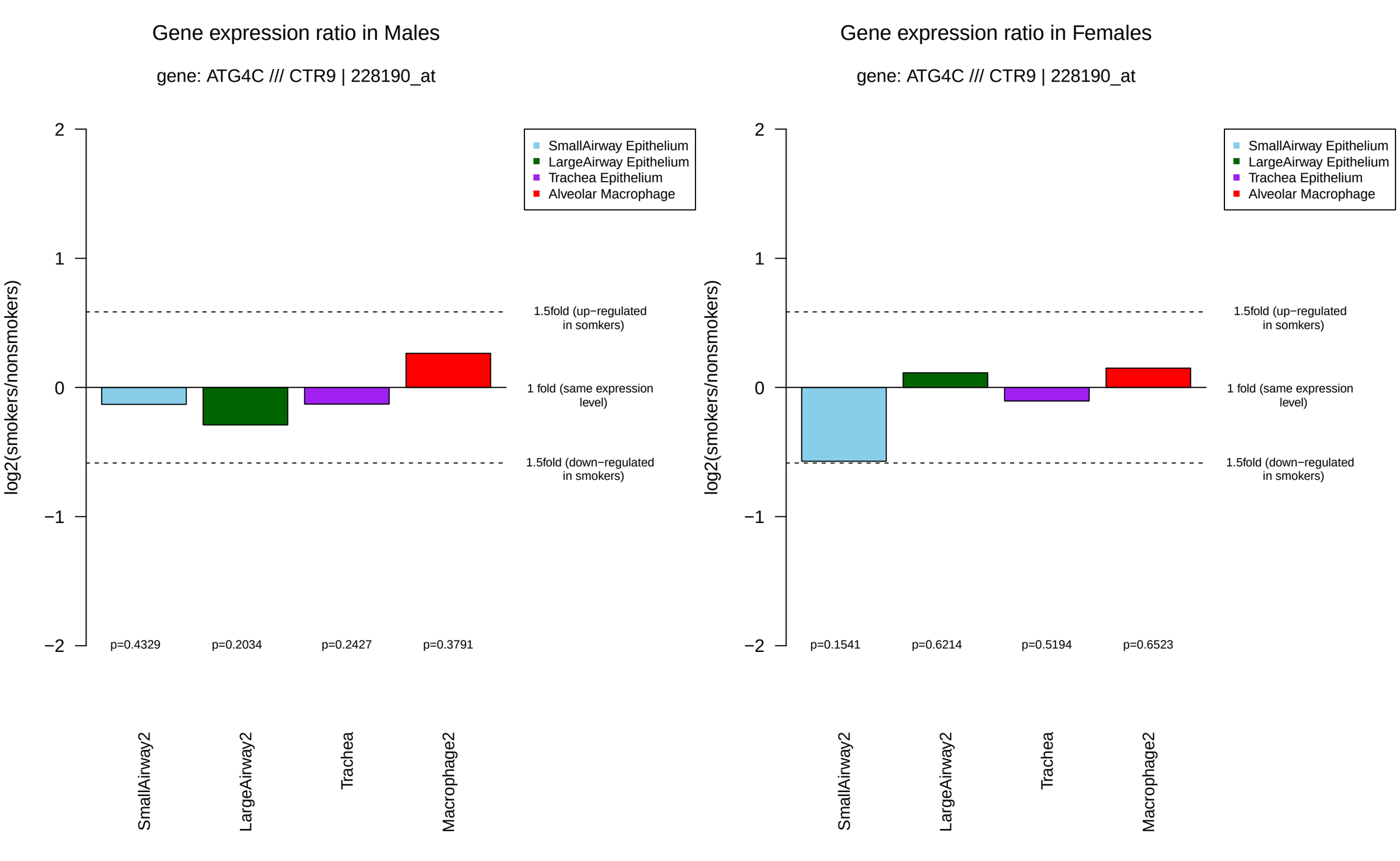
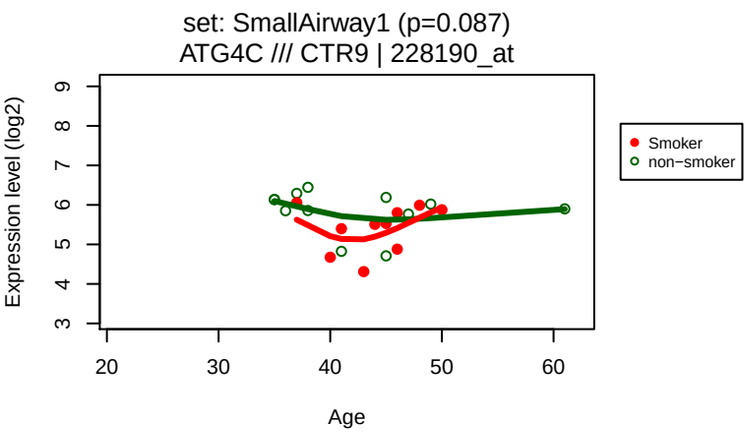
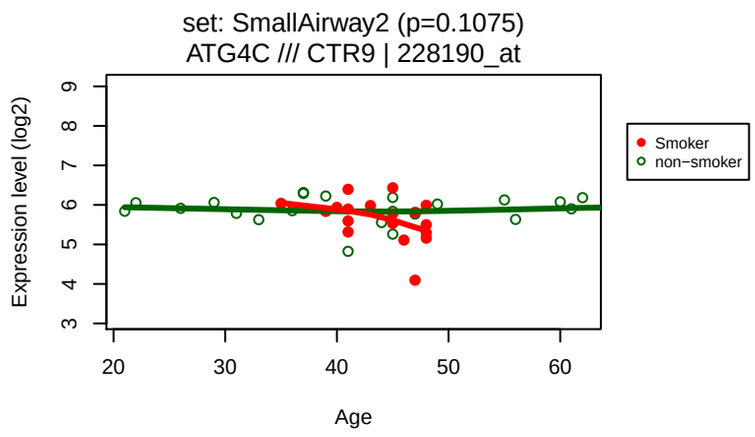
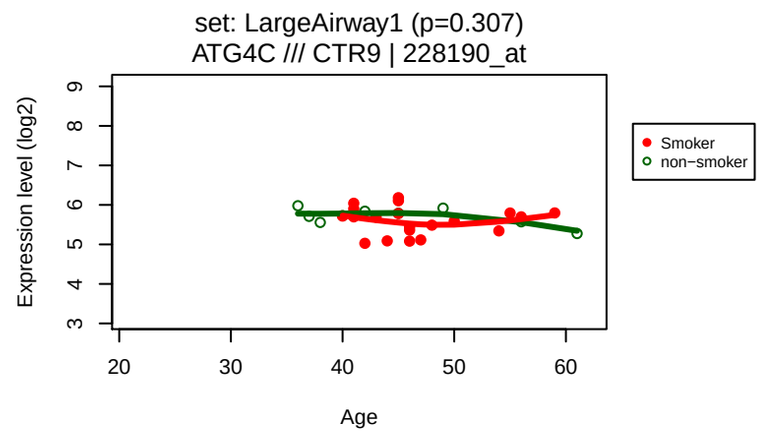

Gene expression of smokers and nonsmokers
gene: ATG4C /// CTR9 | 228190_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.087
p=0.1075
p=0
p=0.307
p=0.2791
p=0.3364
p=0.7038
p=0.2958
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

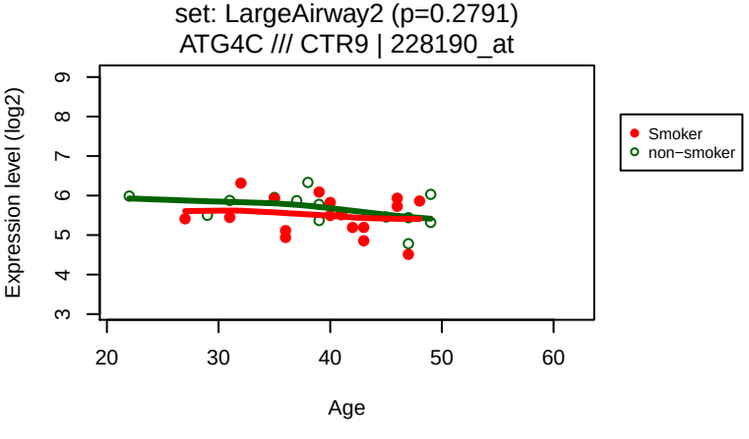

Gene expression ratio (smokers/nonsmokers)
gene: ATG4C /// CTR9 | 228190_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.087
p=0.1075
p=0
p=0.307
p=0.2791
p=0.3364
p=0.7038
p=0.2958
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

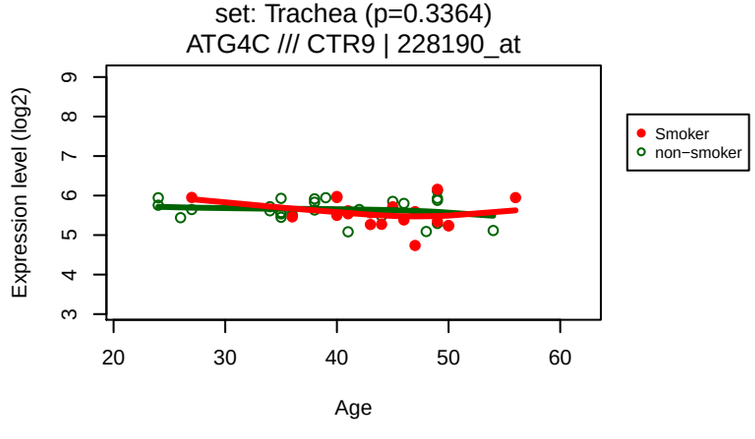

Gene expression ratio in Males
gene: ATG4C /// CTR9 | 228190_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4329
p=0.2034
p=0.2427
p=0.3791
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATG4C /// CTR9 | 228190_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1541
p=0.6214
p=0.5194
p=0.6523
SmallAirway2
LargeAirway2
Trachea
Macrophage2

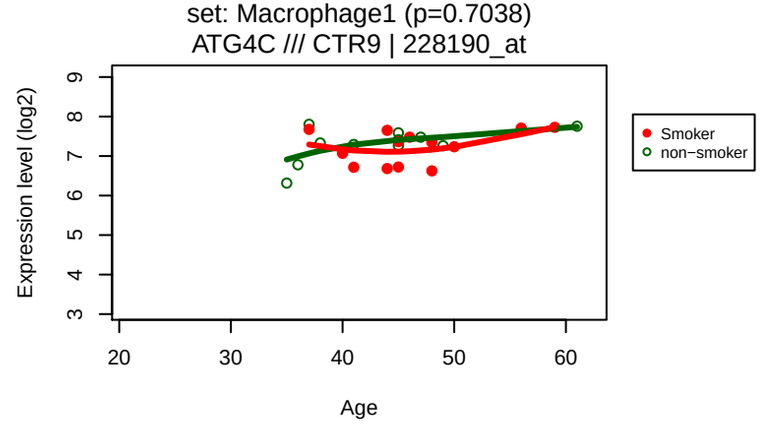

set: SmallAirway1 (p=0.087)
ATG4C /// CTR9 | 228190_at
Expression level (log2)
Age
Smoker
non−smoker

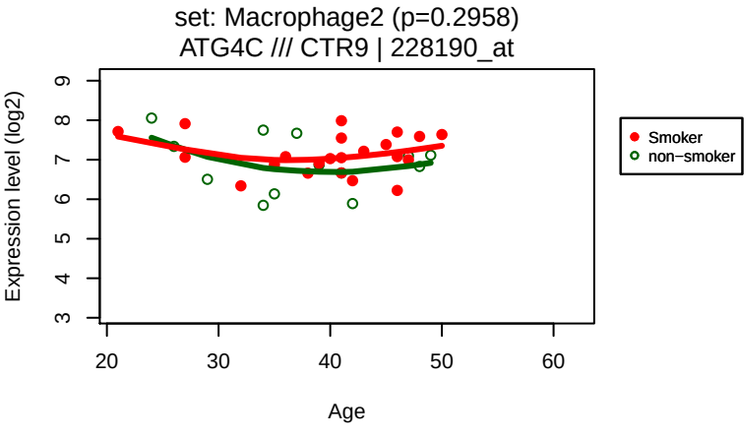

set: SmallAirway2 (p=0.1075)
ATG4C /// CTR9 | 228190_at
Expression level (log2)
Age
Smoker
non−smoker

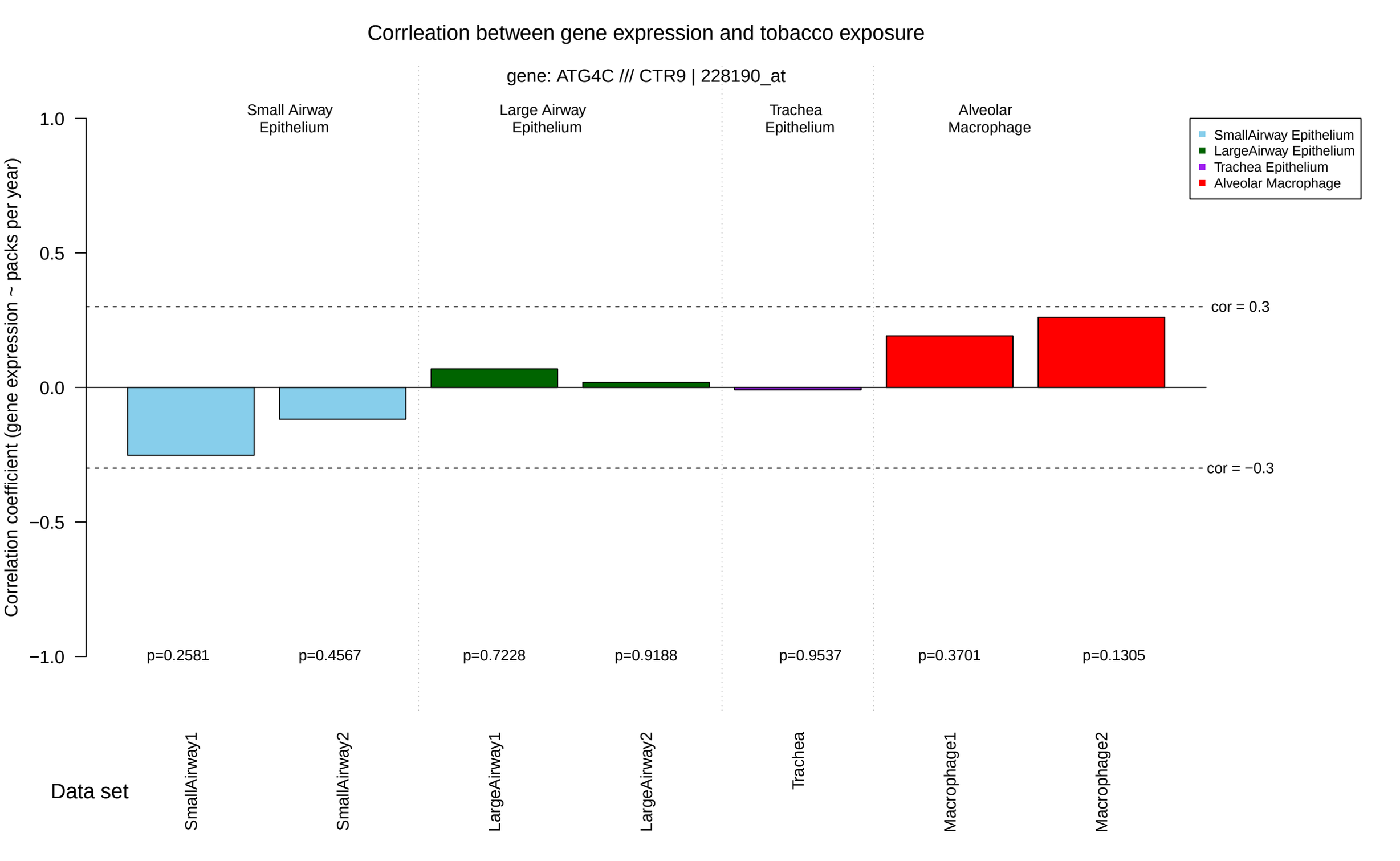

set: LargeAirway1 (p=0.307)
ATG4C /// CTR9 | 228190_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2791)
ATG4C /// CTR9 | 228190_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3364)
ATG4C /// CTR9 | 228190_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7038)
ATG4C /// CTR9 | 228190_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2958)
ATG4C /// CTR9 | 228190_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATG4C /// CTR9 | 228190_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2581
p=0.4567
p=0.7228
p=0.9188
p=0.9537
p=0.3701
p=0.1305
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2